SQL QUERY IMPLEMENTATION FOR E- COMMERCE DATABASE

Swiggy instamart E- commerce platform

Objectives

1. Perform SELECT, WHERE, ORDER BY, and DISTINCT queries.
2. Search products based on price, category, and availability.
3. Retrieve customer and product information.
4. Apply filtering conditions.
5. Generate basic business reports.

1. Basic SQL Queries

1.1 SELECT Query

The SELECT statement is used to retrieve data from a table.

Display all products

```
SQL> SELECT*FROM Product;

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
         1         105
Milk
Aavin                                                  30         50

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
500 ml               23-SEP-26


PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
```

Display selected product details

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, BRAND, PRICE
  2  FROM Product;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE
-------------------------------------------------- ----------
         1
Milk
Aavin                                                    30

         2
Bread
Modern                                                   40

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE
-------------------------------------------------- ----------

         3
Apple
Fresh Farm                                              120

         4
Biscuits

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE
-------------------------------------------------- ----------
Parle                                                    30

         5
Fruit Juice
```

1.2 WHERE Clause

The WHERE clause is used to filter records based on a condition

Find products with price less than 1000

```
SQL> SELECT *
  2  FROM Product
  3  WHERE PRICE < 1000;

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
---------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
         1         105
Milk
Aavin                                                    30         50

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
---------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
500 ml               23-SEP-26


PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
---------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
```

Find products with price greater than 100

```
SQL> SELECT *
  2  FROM Product
  3  WHERE PRICE > 100;

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
         3         101
Apple
Fresh Farm                                             120         20

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
1 kg                 28-SEP-26
```

1.3 ORDER BY

The ORDER BY clause is used to sort the query results.

Products from lowest to highest price

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM Product
  3  ORDER BY PRICE ASC;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Milk
        30

Biscuits
        30

Bread
        40


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Fruit Juice
        60

Apple
       120
```

Products from highest to lowest price

```
SQL> SELECT PRODUCT_NAME, PRICE
  2  FROM Product
  3  ORDER BY PRICE DESC;

PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Apple
       120

Fruit Juice
        60

Bread
        40


PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
Biscuits
        30

Milk
        30
```

1.4 DISTINCT

The DISTINCT keyword is used to display unique values.

Display unique categories

```
SQL>
SQL> SELECT DISTINCT CATEGORY_ID
  2  FROM Product;

CATEGORY_ID
-----------
        105
        104
        101
        103
```

2. Search Products

2.1 Search Products Based on Price

```
SQL> SELECT *
  2  FROM Product
  3  WHERE PRICE BETWEEN 100 AND 400;

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
         3         101
Apple
Fresh Farm                                             120         20

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
1 kg                 28-SEP-26
```

This displays products whose price is between ₹100 and ₹400.

2.2 Search Products Based on Category

```
SQL> SELECT *
  2  FROM Product
  3  WHERE CATEGORY_ID = 101;

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
         3         101
Apple
Fresh Farm                                             120         20

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
1 kg                 28-SEP-26
```

This displays products belonging to category 101.

3.Retrieve Customer Information

Display all customers

```
SQL>
SQL> SELECT CUSTOMER_ID, FIRST_NAME, LAST_NAME, EMAIL
  2  FROM Customer;

CUSTOMER_ID FIRST_NAME
----------- --------------------------------------------------
LAST_NAME
--------------------------------------------------
EMAIL
--------------------------------------------------------------------------------
        101 Rahul
Sharma
rahul@gmail.com

        102 Priya
Kumar
priya@gmail.com
```

Search for a particular customer

```
SQL> SELECT *
  2  FROM Product
  3  WHERE CATEGORY_ID = 101;

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
         3         101
Apple
Fresh Farm                                             120         20

PRODUCT_ID CATEGORY_ID
---------- -----------
PRODUCT_NAME
--------------------------------------------------------------------------------
BRAND                                                PRICE      STOCK
-------------------------------------------------- ---------- ----------
WEIGHT               EXPIRY_DA
-------------------- ---------
IMAGE_URL
--------------------------------------------------------------------------------
1 kg                 28-SEP-26
```

4. Apply Filtering Conditions

Multiple conditions can be combined using AND, OR, and NOT.

4.1 AND Condition

```
SQL>
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM Product
  3  WHERE PRICE < 2000
  4  AND CATEGORY_ID = 2;

no rows selected
```

This displays products that satisfy both conditions.

4.2 OR Condition

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM Product
  3  WHERE CATEGORY_ID = 101
  4  OR CATEGORY_ID = 102;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         3
Apple
       120
```

This displays products belonging to either category 101 or 102.

4.3 IN Operator

```
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM Product
  3  WHERE CATEGORY_ID IN (101, 102, 103);

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         3
Apple
       120

         5
Fruit Juice
        60
```

4.4 BETWEEN Operator

```
SQL>
SQL> SELECT PRODUCT_ID, PRODUCT_NAME, PRICE
  2  FROM Product
  3  WHERE PRICE BETWEEN 100 AND 400;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         3
Apple
       120
```

5. Basic Business Reports

5.1 Product Price Report

```
SQL> SELECT
  2      PRODUCT_ID,
  3      PRODUCT_NAME,
  4      PRICE
  5  FROM Product
  6  ORDER BY PRICE DESC;

PRODUCT_ID
----------
PRODUCT_NAME
--------------------------------------------------------------------------------
     PRICE
----------
         3
Apple
       120

         5
Fruit Juice
```

5.2 Customer Order Report

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.FIRST_NAME || ' ' || c.LAST_NAME AS CUSTOMER_NAME,
  4      o.ORDER_ID,
  5      o.ORDER_DATE,
  6      o.TOTAL_AMOUNT
  7  FROM Customer c
  8  JOIN Orders o
  9      ON c.CUSTOMER_ID = o.CUSTOMER_ID
 10  ORDER BY o.ORDER_DATE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
  ORDER_ID ORDER_DAT TOTAL_AMOUNT
---------- --------- ------------
       105
Vikram Singh
```

5.3 Product Order Report

```
SQL> SELECT
  2      oi.ORDER_ID,
  3      oi.PRODUCT_ID,
  4      oi.QUANTITY,
  5      oi.UNIT_PRICE,
  6      (oi.QUANTITY * oi.UNIT_PRICE) AS SUBTOTAL
  7  FROM Order_Item oi
  8  ORDER BY (oi.QUANTITY * oi.UNIT_PRICE) DESC;

  ORDER_ID PRODUCT_ID   QUANTITY UNIT_PRICE   SUBTOTAL
---------- ---------- ---------- ---------- ----------
      1008          5          2        320        640
      1007          4          3        200        600
      1001          1          3        160        480
      1005          2          4        120        480
      1001          2          2        210        420
      1002          4          2        200        400
      1003          5          1        320        320
      1004          1          2        150        300
      1002          3          3         80        240
      1006          3          2         80        160

10 rows selected.
```

5.4 Customer-wise Total Orders

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.FIRST_NAME || ' ' || c.LAST_NAME AS CUSTOMER_NAME,
  4      COUNT(o.ORDER_ID) AS TOTAL_ORDERS
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.CUSTOMER_ID = o.CUSTOMER_ID
  8  GROUP BY
  9      c.CUSTOMER_ID,
 10      c.FIRST_NAME,
 11      c.LAST_NAME
 12  ORDER BY TOTAL_ORDERS DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_ORDERS
------------
        101
Rahul Sharma
           2

        102
Priya Kumar
           2
```

5.5 Customer-wise Total Purchase Amount

```
SQL> SELECT
  2      c.CUSTOMER_ID,
  3      c.FIRST_NAME || ' ' || c.LAST_NAME AS CUSTOMER_NAME,
  4      SUM(o.TOTAL_AMOUNT) AS TOTAL_PURCHASE
  5  FROM Customer c
  6  JOIN Orders o
  7      ON c.CUSTOMER_ID = o.CUSTOMER_ID
  8  GROUP BY
  9      c.CUSTOMER_ID,
 10      c.FIRST_NAME,
 11      c.LAST_NAME
 12  ORDER BY TOTAL_PURCHASE DESC;

CUSTOMER_ID
-----------
CUSTOMER_NAME
--------------------------------------------------------------------------------
TOTAL_PURCHASE
--------------
        105
Vikram Singh
          1970

        102
Priya Kumar
          1725
```